

Nehme Haidura

Windsor, ON | nehmeheidura@gmail.com | (780) 381-5939

Portfolio: nehme-haidura.github.io

LinkedIn: linkedin.com/in/nehme-haidura

GitHub: github.com/nehme-haidura

Summary

AI and Computer Science researcher with experience in LLM evaluation, software quality assurance, technical support, machine learning, and data analysis. Skilled in AI evaluation workflows, prompt engineering, Python development, NLP methodologies, software testing, and computer vision projects. Experienced in evaluating AI-generated responses for factuality, reasoning quality, and safety while developing research-driven AI systems and technical solutions. Currently completing a Master of Science in Computer Science, Artificial Intelligence Stream, at the University of Windsor.

Technical Skills

Programming:

Python, SQL, Java, HTML, CSS, JavaScript

AI & Machine Learning:

Machine Learning, Deep Learning, LLM Evaluation, NLP, Prompt Engineering, AI Evaluation Frameworks, Computer Vision, OpenCV

Tools & Frameworks:

PyTorch, Scikit-learn, Git, GitHub, Jira, Azure AI, Visual Studio

Data & Analytics:

Data Analysis, Data Visualization, Excel, Reporting, SQL Querying

Technical Support & QA:

Software Testing, QA Analysis, Troubleshooting, Documentation

Education

M.Sc. in Computer Science - AI Stream

University of Windsor, Windsor, ON

May 2025 – August 2026

Junior Data Analyst

NPower Canada, Toronto, ON

May 2024 – August 2024

Bachelor of Computer Science

Lebanese International University, Beirut, Lebanon

September 2018 – May 2021

Professional Experiences

AI Evaluation Specialist

Outlier AI | Remote

2024 – Present

- Analyzed large-scale AI-generated datasets to identify trends, inconsistencies, and quality issues.
- Produced structured evaluation reports and insights to support model performance improvements.
- Worked with cross-functional workflows involving quality analysis, reporting, and data interpretation.
- Applied analytical and problem-solving skills to assess factuality, reasoning quality, and performance metrics.

Graduate Assistant

University of Windsor | Windsor, ON

May 2025 – April 2026

- Assisted students with technical concepts related to cybersecurity, internet technologies, and information systems.
- Supported instructors with grading, academic support, and administrative tasks.
- Helped students troubleshoot technical and course-related issues.
- Collaborated with faculty members to support course delivery and student learning.

Technical Support Specialist

Jiffy Tech Services | Fort McMurray, AB

2023

- Provided technical support by troubleshooting software and system-related issues.
- Diagnosed technical problems and escalated complex incidents when required.
- Maintained clear documentation of issues, troubleshooting steps, and resolutions.
- Assisted customers with technical inquiries and system support.

Quality Assurance Analyst

CARMA | Beirut, Lebanon

2022

- Analyzed software testing results and documented performance trends and defects
- Developed Excel-based tracking and reporting workflows to support operational analysis.
- Collaborated with technical teams to improve system reliability and issue resolution.
- Assisted in data organization, reporting, and process improvement initiatives.

Projects

HLFEval – AI Evaluation Framework

- Designed an AI-driven evaluation framework to analyze and compare generated text quality across multiple performance dimensions.
- Processed and analyzed large datasets using Python-based workflows and automated evaluation methods.
- Generated comparative performance insights and reporting metrics to support analytical decision-making.

ArchVision: From CNNs to Multimodal Scene Classification

- Implemented and evaluated multiple machine learning models for image classification tasks.
- Processed and analyzed large-scale datasets using Python-based tools.

Mixture of Experts (MoE) LLM System

- Designed and tested a system combining multiple AI language models for adaptive text generation.

Research Papers

A Comparative Study on Domain-Specific LLMs: Architecture and Performance

University of Windsor, Windsor, ON

May 2025 – October 2025

- Conducted a comprehensive comparative analysis of 60+ studies on domain-specific large language models (DSLlMs), including BERT, GPT, and T5-based architectures.
- Accepted for presentation at NLPiR 2025, held at Kyushu University in Fukuoka, Japan.
- Accepted / In Press for publication in Springer Lecture Notes in Networks and Systems (LNNS).

Hybrid Metric and LLM-Based Framework for Sentence-Level Evaluation of Generated Text

University of Windsor, Windsor, ON

January 2026 – Present

- Developing a hybrid framework for evaluating LLM-generated text using semantic relevance, fluency, safety, and factuality metrics.
- Implementing LLM-as-a-Judge methodologies, prompt engineering, and automated evaluation pipelines using Python and NLP techniques.

Certifications/Additional Training

IBM Data Analyst, IBM

2024

AI Fundamentals, Microsoft Azure

2024

IT Essentials, Cisco

2021

Routing and Switching, Cisco

2020